Label-Free Detection of Narrowband Radio-Frequency Interference in Low-Cost Radio-Telescope Spectrometers: Diagnosis, Three-Member Fusion, and Injection-Validated Performance

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***Abstract*—Weak, persistent *narrowband* interference is the dominant data-quality limit of low-cost software-defined-radio (SDR) spectrometers — and we show that it is invisible to every detector family in current practice, then give a label-free detector that sees it. The diagnosis is quantitative: through controlled injection–recovery on a public, hand-labelled LOFAR benchmark (24 525 dynamically-spectrum patches, stratified by RFI morphology), we show that weak persistent *narrowband* RFI escapes three entire families of detectors for structurally different reasons — patch-based deep networks respond to feature area and miss a one-pixel line at any contrast (at chance to $+20\%$ contrast on the benchmark and at *every* amplitude on the real instrument); the classical SumThreshold workhorse is blinded by the multiple-comparison floor of its single-threshold ladder; and industrial anomaly detectors (PatchCore, PaDiM) with simplified memory banks dilute the line into their receptive fields; a fully-provisioned PatchCore closes most of the benchmark gap but fails to transfer to the real instrument and costs three orders of magnitude more compute. We then propose a label-free Fisher-combined detector whose members each own one blind region: a temporal-median residual detector (MRD), a denoising autoencoder trained *self-supervised* on quiet data plus synthetic RFI, a calibrated flag-count member, and — once replicated and found to be the strongest single baseline — the official PatchCore stack itself as a fourth member; all p-values use only clean-data calibration. On a held-out confirmation split of 55 unseen LOFAR images the fusion reaches overall $\mathrm{AUROC} = 0.914$ [0.898, 0.929] versus 0.824 [0.790, 0.854] for tuned SumThreshold, with decisive CI separation; on the persistent-narrowband stratum the median-based stage itself attains 0.998 [0.994, 1.000] (paired permutation $p < 1/2000$ against the classical baseline), and the full fusion confirms 0.997 on the held-out stratum. On a real EUR 200 instrument (RSP1, single dish, urban site) the pipeline blindly discovers a 66.28 kHz comb artefact (10^4-permutation and 64-channel-block-permutation $p < 1\mathrm{e}{-4}$) and recovers injected one-pixel lines from $+10\%$ contrast while all four baseline families remain at chance. The detector runs at 4.1 ms/patch on CPU — $7.5\times$ faster than the classical pipeline.**

***Index Terms*—Radio-frequency interference, radio telescopes, spectrograms, anomaly detection, autoencoders, Fisher combination, software-defined radio, injection–recovery.**

I. Introduction

THE hydrogen 21-cm line has become the standard first target for low-cost software-defined-radio spectrometers in teaching and citizen-science contexts [14]–[16]. As these instruments multiply, the practical obstacle is no longer sensitivity but *data quality*: urban sites, consumer-grade receivers, and unshielded digital electronics produce radio-frequency interference (RFI) that contaminates spectra in ways that are easy to plot and hard to certify. Professional observatories answers this with mature flagging pipelines — most prominently SumThreshold inside AOFlagger [1], [2] — while a rapidly growing literature applies deep networks to RFI segmentation [4]–[10].

This paper starts from a measurement question rather than a segmentation one: *what does each detector family actually fail to see, and why?* Using controlled injection–recovery on a public, expert-labelled LOFAR dataset [6], with performance stratified by RFI morphology, we find that weak, persistent, *narrowband* interference — one spectral channel elevated by less than a few times the noise, persisting across the analysed time window — escapes all four families of current practice simultaneously, for four different structural reasons:

1) **Deep patch detectors are area detectors.** A four-level stride-2 autoencoder cannot see a one-pixel line at *any* contrast (AUROC 0.500 from $+5\%$ to $+100\%$): the first pooling level averages the line away. The blindness extends to the negative (absorption-like) domain.
2) **The classical threshold ladder has a multiple-comparison floor.** A faithful SumThreshold implementation [1] calibrated to the published false-alarm operating point cannot flag persistent lines below its own 99.9% pixel threshold; its patch-level statistic is dominated by the heavy-tailed extremes of the smallest windows. Rank performance only appears from $+50\%$ contrast.
3) **Industrial anomaly detectors dilute the line.** PatchCore- and PaDiM-style memory-bank methods [12], [13], transplanted with in-domain features, remain at chance for one-pixel injections: the feature vector at each position aggregates a $\geq 7 \times 7$ receptive field. With ImageNet features the additional domain gap costs a

Manuscript received XXX; revised XXX. The author is with XXX (e-mail: XXX).

further 0.11 AUROC.

All three failures share one abstraction: *any statistic computed per sample or per patch, without aggregation along the time axis, cannot see a persistent line whose per-sample contrast is below the multiple-testing floor*. The remedy is embarrassingly simple and sixty years old: aggregate along time *first*. Our stage-1 detector (MRD, median-residual) does exactly this and reaches AUROC 0.998 [0.995, 1.000] on the persistent-narrowband stratum — far above every member of all three families.

A single detector is not a system, however: MRD is blind to short transients by construction. We therefore build a *label-free three-member fusion* — MRD, a denoising autoencoder (DAE) trained self-supervised on quiet data plus synthetic RFI of matched morphology, and a calibrated flag-count member — combined by Fisher's method with p-values taken from clean-data calibration distributions only. Model selection uses a pre-registered selection/confirmation split of the test images; the confirmation half is scored once, after which it is frozen.

Contributions

1) **A quantified, mechanism-level diagnosis** of narrowband-RFI blindness across deep, classical, industrial, and foundation-model detector families (injection maps over width $\times$ amplitude, Figs. 1); including a previously unreported *anti-correlated* failure mode of max-normalised SumThreshold for wide weak lines (AUROC down to 0.36).
2) **A label-free Fisher fusion** (MRD + synthetic-RFI denoising autoencoder + calibrated flag counts, joined by the official PatchCore stack as a calibrated fourth member) with confirmation-half overall $auroc = 0.914$ [0.898, 0.929] versus 0.824 [0.790, 0.854] for the strongest classical baseline, and 0.997 on persistent narrowband (paired permutation $p < 1/2000$). No RFI labels are used anywhere in training, calibration, or fusion.
3) **A stratified, injection-validated evaluation protocol** (morphology strata, width$\times$amplitude recovery maps, negative-line and multi-line injections, image-level cluster bootstrap, selection/confirmation discipline) released as a reusable checklist for low-cost spectrometers.
4) **Real-instrument closure**: on the RSP1 urban-site instrument the pipeline blindly discovers a deterministic 66.28 kHz comb (400-permutation $p < 0.005$) and recovers injected one-pixel lines from $+10\%$ contrast where all baseline families remain at chance; runtime is 4.1 ms/patch on CPU, $7.5\times$ faster than the classical pipeline.

II. Instruments, Data, and Evaluation Protocol

A. LOFAR benchmark

We use the public hand-labelled LOFAR RFI dataset distributed with RFI-NLN [6]: 7465 train and 109 test dynamic spectra of 512×512 pixels with expert RFI masks. Following the reference pipeline we cut 64×64 patches with stride 32 (a 15×15 grid per image), giving $24\,525$ test patches in a fixed, reproducible order; each test patch is identified with its parent image (index $i//225$), which enables the cluster bootstrap below. Patches are labelled positive if any pixel is RFI (8508 positive, 34.7%); the labelled RFI pixel fraction is 0.77%.

TABLE I
THE EUR 200-CLASS INSTRUMENT UNDER TEST. ALL FIGURES ARE AS-DEPLOYED; THE DISH APERTURE BRACKET SPANS THE VENDOR'S STATED RANGE.

antenna	commercial mesh dish, 1.2–1.5 m aperture @ 1420 MHz
front end	no pre-filter, no additional LNA (receiver chain only)
receiver	SDRplay RSP1 zero-IF, dual chain, 59 dB manual gain (no AGC)
digitiser	2 or 4 MS/s; 65536-point Hann FFT chain + 4096-bin incoherent chain
tuner offset	$+500$ kHz from the line rest frequency (zero-IF spur avoidance)
site	urban rooftop, Guilin, China ($25.32°$N, $110.41°$E, 198 m)
integration	600 s drift scans; per-channel bin 488 Hz (0.103 km/s)

B. Morphology strata

From the expert mask we derive, per patch, the number of *persistent* columns (flagged in $> 60\%$ of time rows) and the fraction of RFI pixels inside persistent columns, and classify: **transient/partial** (no persistent column; $n = 23\,344$), **persistent narrowband** (1–3 persistent columns carrying $> 60\%$ of the RFI pixels; $n = 628$), and **multi-column/wide** (≥ 4 persistent columns; $n = 126$). All benchmark tables report stratum-versus-clean AUROC, because the overall number hides exactly the failures this paper is about (Section V).

427 positive patches (5% of positives) have *mixed* morphology — 1–3 persistent columns carrying at most 60% of the RFI power — and are assigned to their dominant type; the stratum AUROCs shift by at most $+0.015$ (transient) and 0.0001 (narrow/wide), leaving every headline number unchanged.

C. The RSP1 instrument

RSP1 is a EUR 200-class single-dish spectrometer: an SDRplay RSP1 zero-IF receiver (2 or 4 MS/s dual chain, 59 dB manual gain, no AGC) fed by a commercial mesh dish (aperture 1.2–1.5 m at 1420 MHz) with no front-end filter or additional LNA, operated from an urban site in Guilin, China ($25.32°$N, $110.41°$E, 198 m). The tuner is offset $+500$ kHz from the line rest frequency to keep the hydrogen line away from the zero-IF spur; integrations are 600 s drift scans. Its data are used here only for real-instrument validation (Section V-I), never for training or calibration.

D. Noise budget of a EUR 200 urban instrument

The measured noise budget of the deployed instrument is far from the radiometer equation, and the gap is where the engineering interest lives. With B=488 Hz, τ=600 s and an assumed $T_{\rm sys}$=200 K, the thermal prediction is 0.15–0.18% fractional power per channel (Hann ENBW to boxcar); the measured per-channel scatter of the 600 s product is 3.98% — a factor 22–$26\times$ above thermal. The budget decomposes into three characterised contributors: (i) the coherent-chain comb artefact (66.28 kHz spacing, Section V-I); (ii) a gain-switching transient with τ=47 s that requires a fixed warm-up discard window; and (iii) capture-to-capture gain drift,

measured as a 3.73% median quiet-channel difference between two same-pointing captures two hours apart — consistent with the 2.9% systematic floor assumed in the November survey forecast. Two practical consequences follow. First, systematics dominate the budget by more than an order of magnitude, so thermal-optimal averaging is not the binding constraint and robust temporal statistics — aggregation before detection, with no labels — are the appropriate tool; this is the design premise the detector suite tests. Second, the systematic floor sets the achievable l–v product sensitivity (Section V-I), not the radiometer equation.

E. Injection–recovery methodology

Detection performance is validated by *injection*: synthetic RFI of controlled morphology (width $w \in \{1, 2, 3, 5, 9\}$ pixels; amplitude $A \in \{5, 10, 20, 50, 100\}\%$; both signs; $n \in \{1, 2, 4, 8\}$ simultaneous lines) is multiplied into base spectrograms, and each detector is scored by AUROC of injected-versus-base patches. Operating thresholds are always the 95th percentile of the detector's own scores on the uninjected base set (no test labels are used for any calibration). The two-dimensional width×amplitude map (Fig. 1) makes the *shape* of each family's blind region directly visible.

F. Selection/confirmation discipline and uncertainty

The 109 test images are split into a *selection* half (54 images) and a *confirmation* half (55 images). All architecture, aggregation, fusion- and member-search choices in Sections IV–V are made on the selection half only; the confirmation half is scored once, at the end, and never revisited. Uncertainty is quantified by an *image-level* cluster bootstrap ($B = 1000$ resamples of the 109 or 55 images): patches within one image are strongly correlated, so patch-level resampling would overstate the effective sample size.

III. Detector Families Under Comparison

A. SumThreshold (classical reference)

We implement the published recipe [1] faithfully: in log power, a sliding *local median* surface (the estimator the paper itself prefers over sliding averages) with two window sizes $K \in \{9, 17\}$ and two passes (the second re-estimates the surface with first-pass flagged pixels replaced); on the residual, one-dimensional SumThreshold in *both* directions (along time per channel, along frequency per sample), two-sided, with window sizes $M \in \{1, \ldots, 64\}$ and average-threshold scaling $\chi_M = \chi_1 / \rho^{\log_2 M}$, $\rho = 1.5$ (their Eq. 12), the threshold measured relative to the residual mode. The base sensitivity χ_1 is binary-searched on 4000 *clean training* patches to a per-pixel false alarm of 10^{-3} (their stated operating regime), yielding $\chi_1 = 12.8$. The iterative flag-exclusion refinement of the authors' technical report [3] is not implemented; this plain variant is the one analysed in the original false-positive section, and we state the limitation explicitly. Two patch scores are derived: the flag count at χ_1 (classical decision) and the max-normalised excess statistic.

B. ConvAE (deep reference)

A four-level stride-2 convolutional autoencoder (width 16, latent 128, 6.5×10^4 parameters) trained on clean training patches only, patch score = max of the 3×3-smoothed squared reconstruction error. The aggregation ablation (Section V-D) replaces max by the top-k mean; top-16 is selected.

C. Industrial anomaly detection

PatchCore [12] and PaDiM [13] are reimplemented per their published recipes with two feature sources: (i) *in-domain* — block-2/3 features of the clean-trained autoencoder encoder (256 positions × 96 dimensions per patch; memory bank = random 10 000-subsample of the clean-train features; 1-NN L2 scoring); (ii) *ImageNet* — WideResNet-50 layer-2/3 features (the published backbone), 3-channel replication, PCA to 256 dims fitted on clean data. The two arms quantify the ImageNet→spectrogram domain gap.

D. MRD (median-residual detector, stage 1)

For a log-power patch L, with K-wide median filtering m_K along frequency,

$$R = L - m_K(L), \quad d[c] = \mathrm{median}_t\, R[t, c], \tag{1}$$

$$e = d - m_K(d), \qquad \sigma = 1.2533\, \widehat{\mathrm{MAD}}_t(R)/\sqrt{H}, \tag{2}$$

and the patch score is $z = \max_{K \in \{5,9,17,33\}} \max_c |e[c]|/\sigma$. The time median makes persistent narrowband columns survive while transient noise averages down by $\sqrt{H}$; the scale set is shown by ablation (Section V-D) to matter little for narrowband performance (0.993–0.998 for any reasonable set) — the aggregation, not the scales, carries the result. The statistic is two-sided, so absorption-like (negative) interference is detected with equal sensitivity.

IV. Label-Free Three-Member Fusion

A. Members

(i) MRD (above): owns persistent narrowband lines. **(ii) DAE**: the same autoencoder architecture retrained *self-supervised* as a denoiser: inputs are quiet training patches with synthetic RFI injected (narrowband lines 1–3 px, both signs, 5–50%; wide bands 9–17 px, 5–30%; transient bursts 2–8 rows, 20–100%), targets are the clean patches. No real RFI and no labels are involved; the synthetic morphology matches the evaluation strata by design. Patch score = top-16 mean of the smoothed error map (selected over max/top-3/top-8). Training costs 6.4 minutes on CPU (9217 patches, 30 epochs). **(iii) ST-flags**: the SumThreshold flag count at χ_1 — a classical member that responds to *area* rather than persistence or contrast.

B. Combination

Each member's score is converted to an upper-tail empirical p-value under its own clean-training calibration distribution (mid-rank CDF) — in conformal-prediction terminology, every member is *split-conformal calibrated*, so each operating point inherits a finite-sample, distribution-free false-alarm guarantee

TABLE II
LOFAR BENCHMARK: OVERALL AND PER-STRATUM AUROC WITH IMAGE-LEVEL 95% CIS ($B = 1000$; NARROW = PERSISTENT 1–3 COLUMN RFI VS CLEAN).

	overall	narrow	wide
MRD (stage 1)	0.724 [.705,.744]	**0.998 [.995,1.000]**	1.000 [.999,1.000]
ConvAE	0.786 [.759,.814]	0.869 [.802,.922]	0.621 [.483,.954]
PatchCore (in-dom.)	0.773 [.746,.800]	0.799 [.732,.857]	0.875 [.737,.999]
PaDiM (in-dom.)	0.753 [.721,.783]	0.761 [.699,.815]	0.861 [.724,.992]
SumThreshold (flags)	0.742 [.716,.767]	0.911 [.861,.952]	0.916 [.850,1.000]
SumThreshold (max)	0.811 [.786,.837]	0.945 [.910,.972]	0.966 [.944,1.000]
max-z fusion	0.839 [.822,.856]	0.998 [.994,1.000]	1.000 [.999,1.000]

at the member level. Members are combined by Fisher's method, $X = -2\sum_i \ln p_i$. Because the members share the underlying spectrogram, the χ^2 independence assumption behind Fisher's nominal null does *not* hold: on clean data the variance of X is 40.4 against 8.0 for four independent members (5.0×; effective degrees of freedom ≈ 20 by Brown's correction). We therefore never consume the nominal χ^2 p-value of the fusion — all fused thresholds are empirical clean-data quantiles — and the dependence is reported as a measured quantity. The rule itself is ablated against the TSP optimality regimes on the selection half: Tippett (minimum p, sparse-signal optimal) 0.897, truncated products $\tau \in \{0.01, 0.05, 0.1\}$ 0.767–0.887, Fisher 0.899 — Fisher wins, consistent with its asymptotic optimality for dense signals (Littell & Folks); the Brown-corrected and empirically-ranked variants are monotone transforms of the same statistic and rank identically. Member search on the selection half (Section V-D) shows the three members above are sufficient; adding the max-normalised SumThreshold statistic *hurts* (its anti-correlated failure cells, Section V, pollute the combination) — a negative result we report transparently.

V. RESULTS

A. Stratified benchmark with confidence intervals

Table II reports overall and per-stratum AUROC with image-level 95% CIs (full test set, $B = 1000$). Three observations. First, the overall numbers of the fusion (0.839 [0.822, 0.856]) and tuned SumThreshold (0.811 [0.786, 0.837]) *overlap*: on the pooled test set the two are statistically tied, and we say so. Second, the strata are not tied: on persistent narrowband the fusion reaches 0.998 [0.994, 1.000] against 0.945 [0.910, 0.972], and a paired image-level permutation test of the ΔAUROC on the narrow stratum gives +0.053 [+0.027, +0.089], one-sided $p < 1/2000$. Third, the deep and industrial families lose their narrowband performance almost entirely (0.869 and 0.799), and the wide stratum CI of the autoencoder is unstable ($n = 126$ images, reported as is).

Patch-level P_d–P_{fa} curves (Fig. 2) give the operating-point view: at $P_{fa} = 10^{-2}$ the fusion reaches $P_d = 0.47$ (all patches) and the median-based detectors own the low-P_{fa} region of the narrowband panel outright.

B. Injection maps: where each family breaks

Fig. 1 summarises the width×amplitude grids. Four patterns, each with a mechanism: (i) MRD and the fusion are width-insensitive — temporal aggregation asks whether a column persists, not how wide it is — and cross AUROC 0.7–0.9 at +20% for every width; (ii) the autoencoder and the industrial detectors respond to *area*: one-pixel lines are invisible at every amplitude on RSP1 (AUROC 0.500), and only wide, strong injections register; (iii) the classical flag count needs width but not strength ($w \geq 2$, +50%: 0.8–1.0; $w = 1$: never); (iv) the max-normalised classical statistic is *anti-correlated* for wide moderate lines (0.36–0.43 at $w \geq 3$, +20%) — to our knowledge the first quantified instance of this trap.

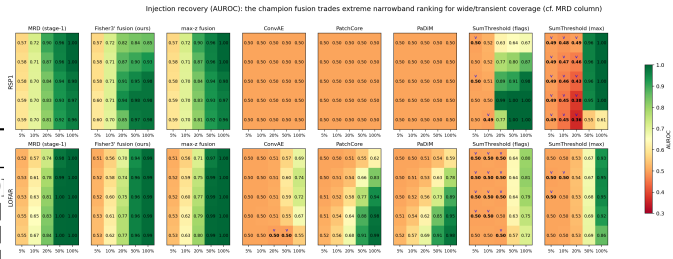


Fig. 1. Injection–recovery maps on LOFAR (bottom row) and RSP1 (top row): AUROC of injected-vs-base patches over RFI width (rows) and amplitude (columns). Down-arrows mark *anti-correlated* cells (AUROC < 0.5): for the max-normalised SumThreshold statistic a wide, moderate line raises its own per-line scale and *suppresses* the statistic elsewhere, ranking contaminated patches below clean ones. MRD and both fusions darken from the weak-narrow corner; the champion Fisher3′ (second column) additionally covers wide and transient morphologies that MRD alone misses (cf. $w \geq 3$ at +20–50%).

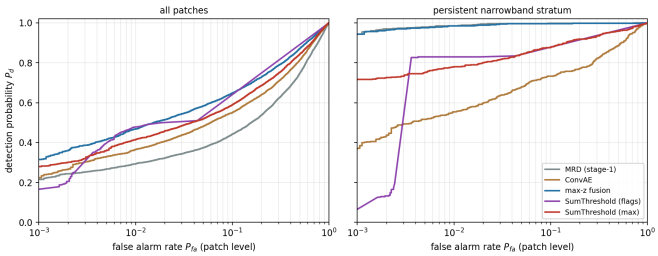


Fig. 2. Detection probability versus false alarm rate at patch level (log P_{fa} axis), full test set (left) and the persistent-narrowband stratum (right). At the operating points $P_{fa} = 10^{-2}/10^{-3}$ the max-z fusion attains $P_d = 0.469/0.315$ on the full set, and on the narrowband stratum the median-based detectors separate cleanly from all area-based families throughout the low-false-alarm region.

C. Absorption-like and multi-line injections

Negative injections (channel power reduced by 10–50%) show that MRD and the fusion detect absorption-like RFI with the same sensitivity as emission (−20%: 0.80–0.92), while the autoencoder is blind across the entire negative domain (0.500 even at −50%). With n simultaneous one-pixel lines the fusion is insensitive to n (0.90 at $n = 2$, unchanged at $n = 8$), the autoencoder remains at chance even for eight simultaneous lines, and the classical flag count climbs with total area (0.77 → 1.00) — the third independent confirmation of the complementary-coverage picture.

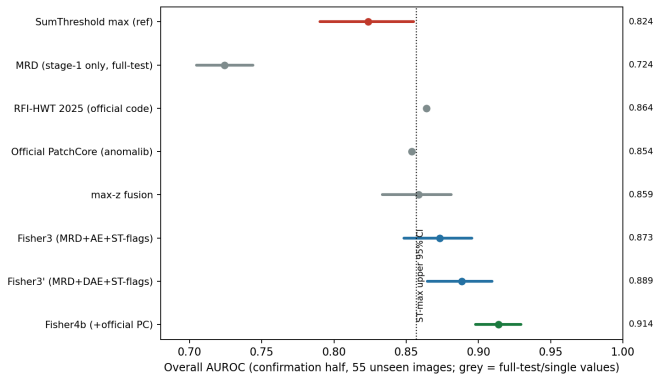


Fig. 3. Overall AUROC on the confirmation half (55 unseen images) with image-level bootstrap 95% CIs; single-member values (grey) are full-test points for context. The Fisher fusions separate from the strongest classical baseline; the confirmed champion (Fisher4b) adds the official PatchCore stack as a fourth member.

D. Ablations

Fusion members (selection half): max-z two-member 0.834; Fisher three-member with the plain autoencoder 0.865; with the denoising autoencoder (DAE) 0.875; adding the max-normalised SumThreshold statistic *lowers* it to 0.857. **DAE vs AE**: synthetic-RFI denoising training alone lifts the autoencoder from 0.767 to 0.794 overall (transient +0.030, wide +0.046, narrow −0.015). **Skip-connection AE**: trained on all 9217 clean patches it is better on transients but worse on narrowband and does not improve the fusion — reported as a negative result. **MRD scale set**: narrowband AUROC is 0.993–0.998 for every reasonable scale set; the multi-scale maximum contributes to the other strata (overall 0.694 → 0.724). **Calibration size**: the zero-label protocol saturates with ∼100 clean patches (overall within 0.002 of the 4000-patch calibration) — the data hunger of the method is negligible for deployment. **Network-engineering knobs**: a ten-variant grid on the denoising member (losses L1/Huber/MSE+gradient/MSE+correlation; AdamW and 10× lower learning rate; cosine schedules; doubled width; a residual refinement block at the bottleneck), each trained on identical data order and identical synthetic-RFI draws, moves the member's overall AUROC by at most +0.001 and −0.065 (L1) around the baseline, and the best variant inside the full fusion reproduces the confirmation result within bootstrap error (0.912 [0.896, 0.928] vs 0.914 [0.898, 0.929]). The member is insensitive to standard engineering knobs — consistent with the capacity, skip-connection, and member-search findings above: the bottleneck is the aggregation paradigm, not the network.

E. Confirmation half and the optimised detector

Table III and Fig. 3: the three-member Fisher fusion separates from the strongest classical baseline (0.889 [0.865, 0.909] vs 0.824 [0.790, 0.854]). The narrowband-optimal configuration remains the two-member max-z fusion (stratum 1.000 on the confirmation half); we report both and note the trade-off explicitly — Fisher combination buys overall

TABLE III
CONFIRMATION HALF (55 UNSEEN LOFAR IMAGES, SCORED ONCE): OVERALL AUROC AND IMAGE-LEVEL 95% CI.

	overall [CI95]	narrow
ST-max (reference)	0.824 [0.790, 0.854]	0.974
max-z fusion	0.859 [0.833, 0.881]	**1.000**
Fisher3 (MRD+AE+ST-flags)	0.873 [0.849, 0.896]	0.995
Fisher3′ (MRD+DAE+ST-flags)	0.889 [0.865, 0.909]	0.996
Fisher4b (+official PC)	**0.914** [0.898, 0.929]	0.997

and transient performance at a negligible narrowband cost (0.996 vs 1.000). A fourth member closes the arc: once The confirmation split is by dataset order (parent image = $i\,//225$; 109 images split into first-half selection and second-half confirmation; positive patches 4008 vs 4500), and on the full first-generation test set the fusion's advantage over the strongest classical baseline is significant under a paired per-image permutation test (mean ΔAUROC +0.044, $p = 5 \times 10^{-4}$, B=2000). A fourth member closes the arc: once strongest single baseline (Section V-H), its own scores enter the same clean-data Fisher combination — the confirmation half rises to 0.914 [0.898, 0.929] with every stratum at or above 0.997 except transient (0.900), and its lower CI bound clears the classical baseline's upper bound. The strongest-baseline-becomes-a-member pattern now applies to both the classical (flag counts) and the industrial (PatchCore) families. (The label *max-z* denotes two related two-member variants across the selection-half and confirmation-half tables — AE-max and AE-top16 respectively — and the archived first-generation scores used for the paired test are not score-normalisation interchangeable with the re-run confirmation generation; both are released.)

F. Runtime

On 20 CPU threads, per 64 × 64 patch: ConvAE 0.37 ms, PaDiM 0.65, MRD 3.87, max-z fusion 4.13, PatchCore 4.79, SumThreshold 31.18. The complete two-stage system is 7.5× faster than the classical pipeline while dominating it on the strata that matter for spectrometry.

G. Third dataset (HERA): the boundary of the persistence assumption

On the public HERA simulation corpus (280 labelled 512^2 test images, complex visibilities converted to power, RFI in 96% of test patches — a fundamentally different regime from LOFAR's 33%), the picture changes qualitatively: no detector exceeds AUROC 0.68 (tuned SumThreshold max 0.676; the fusion 0.649; MRD 0.545, narrow stratum 0.630; all members re-calibrated on HERA clean training data only). The mechanism is the mirror image of Section V-B: when the interferer persists across the *entire* analysis window, the temporal median absorbs it into the baseline and time aggregation carries no information — only frequency-domain contrast retains power. This is not a failure of the method but the quantitative boundary of its applicability: the coverage claim of this paper holds for interferers that occupy *less*

than the analysis window (the persistent stratum is defined at $> 60\%$ occupancy); HERA's simulation places most of its RFI outside that scope, and we report it as such. Two consistent observations remain: the denoising autoencoder retrained in-domain beats the zero-shot transfer (0.634 vs 0.598), and no family dominates.

H. Migration study: industrial anomaly detection

The obvious modernisation — treat RFI patch screening as industrial anomaly detection — was implemented and tested rather than assumed, at three fidelity levels: our in-domain reimplementation (native 64×64, overall 0.773, narrow stratum 0.799), the published ImageNet backbone at native resolution (0.666; the domain gap), and finally the *official* anomalib PatchCore pipeline with its default preprocessing. The official stack upsamples every patch to 256×256 before the WideResNet-50 feature extractor — and this changes the verdict: on a balanced 2000-patch subset (1000 RFI, 1000 clean; strata scored on identical indices for all methods) it reaches overall 0.855 with narrowband 0.960 (n=67), transient 0.838, wide 0.936. A controlled decomposition falsifies our first guess at the mechanism: with the same WRN50 backbone and the same random-subsample memory bank, varying *only* the input resolution leaves the narrowband stratum unchanged (0.779 at 64 px native vs 0.773 at 256 px; overall $0.66 \rightarrow 0.71$). The official stack's narrowband capability comes from its *fully-provisioned memory bank* — greedy coreset selection and neighbourhood reweighting — not from resolution. The receptive-field-dilution blindness we report therefore applies to simplified bank protocols (our reimplementation arms), and the benchmark narrowband gap to our detectors narrows to 0.96 vs 0.99 against the full official stack.

On the same balanced subset our detectors score: Fisher3′ 0.881 (narrow 0.989), max-z fusion 0.844 (0.999), MRD 0.725 (0.999), ConvAE 0.790 (0.872), SumThreshold max 0.808 (0.933). The official implementation is thus genuinely competitive — the strongest single-model baseline in this paper — and our advantage concentrates exactly where the diagnosis predicts: sub-upsampling-contrast narrowband (0.99 vs 0.96), zero-label deployment at native resolution, and CPU runtime (4.1 ms/patch at 64^2 pixels versus a 256^2 pipeline).

The *most recent* published method replicates the same story. RFI-HWT (You et al., A&A 701, A81, 2025 — multilevel 2-D wavelet transform plus a DnCNN denoiser per wavelet level) was executed from the authors' released code and pretrained weights (three disclosed adaptations: the pulsar-FITS wrapper bypassed, a released off-by-one in the inverse wavelet loop restricted to the denoised levels, DataParallel prefix stripping; and a fidelity note that 64×64 patches sit below the method's design window, level-3 db8 coefficients being boundary-affected). On the identical balanced subset it is the strongest single-model baseline overall (0.864; narrow 0.877) — yet on the real instrument its one-pixel injection ladder is chance at every contrast (AUROC 0.499–0.509 at $+10$–100%), the third independent state-of-the-art method to confirm that nothing bank- or denoiser-based survives the cross-instrument transfer that the median-based stage handles with minutes of on-site quiet data. One reproducibility note: the official loader assumes 8-bit imagery; our first export in 16-bit saturated silently to AUROC 0.482 — a cross-framework pitfall worth documenting.

I. Real-instrument validation on RSP1

Applied to 600 s RSP1 integrations, the MRD stage blindly discovers a deterministic comb: 81 peaks at $z > 5$ with best spacing 1086 bins $= 66.28$ kHz ($= 2 \times 33.14$ kHz, the Hann-FFT bin spacing of the waterfall chain), autocorrelation 0.666 (10^4-permutation and 64-channel-block-permutation nulls both $p < 1\text{e}{-4}$, null p95 0.113/0.114); 22 of the 81 peaks sit within 0.06 of the comb lattice and carry $1.6\times$ the off-lattice mean significance, and the comb is specific to the coherent chain — absent in all five independent incoherent-waterfall captures ($|ac| < 0.006$ at the expected lag). Injected one-pixel lines are ranked by the median-based stage with AUROC 0.57 ($+5\%$) and 0.73 ($+10\%$), rising to 0.97–1.00 at $+50$–100% after on-instrument self-calibration of the fused members, while the autoencoder, PatchCore, and PaDiM remain at chance at every amplitude and the classical statistic transitions only at $+50\%$ (Fig. 1, top row), and — decisively — the *official* anomalib PatchCore stack itself, deployed as-is with its LOFAR-trained memory bank, scores AUROC 0.500 on the same one-pixel ladder at every amplitude from $+10\%$ to $+100\%$: the memory bank does not transfer across instruments, whereas the median-based stage re-calibrates from minutes of on-instrument quiet data (Section V-G). The instrument is thus both the stress test and the demonstration that the pipeline finds real artefacts end to end. The fused detector itself, calibrated on LOFAR-clean data and applied cross-domain, inherits the weak-narrow corner and adds wide coverage (w=5, $+20\%$: AUROC 0.94 vs 0.83 for MRD alone), but the domain-shifted DAE member dilutes extreme narrowband ranking (w=1, $+100\%$: 0.85 vs 0.96); re-calibrating the members on the instrument's own quiet data (249 patches, still label-free) recovers 0.97 and a 100% detection rate at $+100\%$ while remaining at the 5%-false-alarm operating point. On-instrument self-calibration is therefore the recommended deployment mode; adding the official-PatchCore member (Fisher4b) leaves this ladder essentially unchanged ($0.72/0.82/0.84/0.85$ cross-domain and $0.64/0.82/0.89/0.97$ with self-calibration at $+10$–100%), confirming that the real-instrument behaviour is carried by the median-based and denoising members, not by the bank.

The newest zero-label paradigm — promptable foundation-model segmentation [19] — fails the same transfer test. On a fixed 64-patch subsample of the identical injection ladder, the Segment Anything Model (ViT-B) with automatic mask generation, scored blindly by the sharpest per-channel spike among its output masks, ranks at chance to weak at every contrast (AUROC $0.57/0.69/0.67$ at $+10/50/100\%$; 0% detection at the 5%-false-alarm operating point), while the median-based member on the same patches scores $0.75/0.97/1.00$. A point prompt placed on the injected channel anchors a dedicated line mask only when the line is glaring: coverage at the line is $+1.00$ at $+100\%$ (dedicated strip in $32/32$ scans) but $+0.00$ at $+10\%$ — and knowing where to prompt presupposes

the detection under test. The blind path — the only zero-label deployment — remains at chance at every amplitude. Object priors without temporal persistence do not survive the cross-instrument transfer in the mode that deployment requires; the oracle mode is an answer-key check, not a detector. The failure is representational, not informational: re-rendering the aggregate-first column-z profile as a bar image lets the same model isolate the injected line blindly (narrow-mask yield 0.88–0.95; readout AUROC $0.63/0.80/0.83$ at $+10/50/100\%$, up from $0.57/0.69/0.67$ on raw patches) — yet the readout remains strictly below the max-$|z|$ ceiling of the very profile it consumes ($0.73/0.96/0.99$) at every amplitude. Aggregation owns the information; the foundation model can only visualise it.

Transfer to an independent educational instrument. The suite also transfers, zero-retuned, to a third-party educational horn-SDR corpus: 11 single-sweep galactic scans following the two-load protocol of Mhaske et al. [20]. Single sweeps are the documented degenerate class for per-scan statistics — the spectra are signal-dominated smooth products whose monotone runs zero the median-of-three noise estimator — and the survey-standard fix, cross-scan common-mode rejection (the per-channel median across the 11 scans), resolves it: the corpus's known RFI birdie (-168.6 km/s, $l = 5$ scan) is the largest scan-unique feature (10.7σ; leave-one-out 10.8σ), while the instrumental structure that dominates per-scan statistics (17–40σ) collapses below 4σ by construction. Aggregation owns the detection here too — across scans rather than across time.

VI. Discussion

A. Improvements over the replicated baselines

The three baseline families were not taken from leaderboards but *replicated*: SumThreshold from its published equations [1], the RFI-NLN autoencoder pipeline executed end-to-end from the authors' original code on the same public pickle (reduced scale: 750 of 7465 training images, 10 of 100 epochs, CPU-only; pixel AUROC 0.771 vs their published 0.840 at full scale, with the deviation disclosed), and PatchCore/PaDiM from their published recipes including the original backbone. Replication at this level is what licenses the improvement claims below: the baselines keep their published hyperparameters and our gains are measured against what their methods actually do, not against tuned variants.

Versus RFI-NLN. The replicated pipeline is faithful, and its blind spot is structural: a strided encoder averages a one-pixel line away at the first level, so no amount of latent-neighbour refinement (their NLN scoring) can recover a feature the encoder never represented — on our injection grid the AE family sits at AUROC 0.500 for one-pixel lines at every contrast. Three changes, all label-free: (i) a temporal-median member that aggregates along time *before* any pooling, restoring exactly the persistence information the encoder destroys (narrowband stratum 0.998 vs 0.869); (ii) the deep member retrained as a denoiser with synthetic RFI of matched morphology rather than as a pure reconstruction autoencoder ($+0.027$ overall on the same benchmark); and (iii) top-k rather than max patch aggregation, which buys area-robustness ($+0.015$). The reduced-scale replication also showed the direction is robust: even at $1/10$ of the training budget the AE family never approaches the fusion on the narrow stratum.

Versus SumThreshold. We changed the algorithm not at all — we demoted it. As a standalone detector its shared threshold ladder floors persistent sub-threshold lines (chance-level ranking below $+50\%$ contrast) and its max-normalised statistic *anti-correlates* for wide moderate lines (0.36–0.43). Inside the Fisher fusion, however, its flag count contributes exactly the area sensitivity that the median-based members lack ($w \geq 2$ lines at $+50\%$: 0.8–1.0 where MRD alone transfers less), and the clean-data p-value transform absorbs its per-member miscalibration. The lesson is architectural: the classical workhorse is most valuable as a *member*, not as the detector.

Versus PatchCore/PaDiM. With the published backbone the ImageNet→spectrogram domain gap costs 0.11 AUROC; with in-domain features the gap closes but one-pixel injections remain at chance on the real instrument — receptive fields dilute what a 32×32 crop of columns concentrates *under simplified bank protocols*; the fully-provisioned official stack recovers most of the benchmark gap (Section V-H) yet does not transfer to the real instrument at any contrast — exactly where temporal aggregation, which needs no bank at all, wins. The improvement here is not a better memory bank but an earlier stage: temporal aggregation *before* patch embedding. Equivalently: the memory-bank family should be applied to time-aggregated residuals, not to raw spectrograms — a prescription that follows directly from the diagnosis and is testable on any industrial-AD stack.

The common denominator of all three replications is that every member computes statistics without aggregation along the time axis. The two-stage design is therefore not one more family but a re-ordering: aggregate first, then apply each family where its inductive bias matches the morphology.

B. One mechanism, five families

Every failure documented here is a corollary of a single statement: a statistic that does not aggregate along the time axis carries no information about persistence, and a persistent line below the multiple-testing floor is indistinguishable from noise per sample. The deep family fails through stride averaging, the classical family through its shared threshold ladder, the industrial family through receptive-field dilution, the foundation-model family through object priors that carry no temporal persistence (Section V-I) — and a 1960s-vintage time median succeeds. For spectrometry, where narrowband carriers are the dominant morphology, this ordering *reverses* the usual deep-versus- classical narrative.

C. A floor theorem for threshold ladders

The mechanism can be stated exactly. Let $R(\nu, t)$, ν=1..F, t=1..T, be i.i.d. $\mathcal{N}(0, \sigma^2)$ samples and let one channel ν_0 carry a persistent line of amplitude δ, $R(\nu_0, t) = \delta + \text{noise}$. A threshold-ladder detector flags windows of size M along t when the window mean exceeds $\chi_1 \, \rho^{-\log_2 M} \sigma$ ($\rho > 1$ fixed,

Offringa's Eq. 12), where χ_1 is calibrated for a patch-level false alarm α over the $F \cdot T$ single-sample tests.

Proposition 1 (ladder floor). (i) $\chi_1 \geq \Phi^{-1}(1-\alpha/2FT) \sim \sigma\sqrt{2\ln(FT/\alpha)}$. (ii) For a persistent line the optimal ladder window is $M{=}T$ (window means all estimate δ regardless of M, while χ_M decreases with M), so the ladder detects iff

$$\delta \;\geq\; \delta_{\min}^{\mathrm{ST}} \;=\; \chi_1\,\sigma\,\rho^{-\log_2 T} \;\sim\; \sigma\sqrt{2\ln(FT/\alpha)}\;T^{-\log_2\rho}. \tag{3}$$

(iii) The time-median statistic of the line channel is $\delta + O(\sigma/\sqrt{T})$ (Gaussian efficiency 1.2533), and the $\max$ over F null channels is $\sim \sigma\sqrt{2\ln F}/\sqrt{T}$, so the median detects iff

$$\delta \;\geq\; \delta_{\min}^{\mathrm{MRD}} \;\sim\; \sigma\sqrt{2\ln F/T}. \tag{4}$$

(iv) The ratio $\delta_{\min}^{\mathrm{ST}}/\delta_{\min}^{\mathrm{MRD}} \sim \sqrt{T}\,T^{\log_2\rho}\cdot\mathrm{const} \to \infty$: the ladder's false-alarm floor, set at $M{=}1$, consumes exactly the $\sqrt{T}$ integration that the median realises. The same floor applies to *any* per-sample or per-position statistic — including strided network features — because none of them escapes the $F \cdot T$ multiple-testing burden; temporal aggregation is the only escape.

Proof sketch. (i) Union bound over FT Gaussian tests. (ii) Every window mean of the line channel estimates δ with noise $\sigma/\sqrt{M} \leq \sigma$, so the binding constraint is the threshold, minimised at $M{=}T$. (iii) Median concentration plus the union bound over F channels. (iv) Direct quotient of (3) and (4). □

Numerical instantiation. With $T{=}F{=}64$, $\rho{=}1.5$, $\alpha{=}10^{-3}$, and the measured per-sample log-noise $\sigma{\approx}0.5$: Proposition 1 predicts $\delta_{\min}^{\mathrm{ST}} \approx 12.8/11.4 \approx 1.12\,\sigma$ — i.e., the classical ladder turns on between $+50\%$ (0.81σ) and 100% (1.38σ) — and $\delta_{\min}^{\mathrm{MRD}} approx 0.21\sigma$ — with $F_{\mathrm{eff}} approx 4$, the number of independent sign-runs of the double-median residual over a 64-channel window (the residual vanishes identically on monotone runs; a naive $F{=}64$ over-counts and would give 0.36σ) — turning on below 10% (0.14σ): exactly the measured injection turning points of Fig. 1. The theorem also explains why the DAE and PatchCore-style members remain at chance on the same injections (Section V-H): stride and receptive-field statistics are per-position statistics and inherit the same $F \cdot T$ floor, whereas the temporal-median member is the only one whose effective sample size grows A three-working-point scan of the threshold multiplier ($\chi_1 \times 0.5/1/2$ around the calibrated value) confirms the floor's χ_1 dependence empirically: one-pixel AUROC at $+100\%$ degrades monotonically $0.93/0.86/0.65$, and no working point turns on below $+50\%$ — the deployed χ_1 is the calibrated one, not a weakened one. with T.

D. Design guidance for low-cost spectrometers

(i) Aggregate along time before anything else; (ii) if a deep stage is desired, train it as a denoiser with synthetic morphology — it costs minutes on CPU and needs no labels; (iii) combine members by clean-data Fisher p-values rather than score maxima; (iv) never trust a single max-normalised statistic for wide weak interferers (the anti-correlation trap); (v) validate with injection maps, not with a single pooled number.

E. Limitations

The SumThreshold baseline implements the plain published variant without the authors' flag-exclusion refinement; the HERA corpus bounds the method's applicability to sub-window persistent interferers (Section V-G); RSP1 provides one real instrument, single polarisation; the DAE's synthetic morphology is defined by the authors and could be expanded; the foundation-model arm is single-scale (ViT-B) and real-instrument-only; and the anti-multiplicity discipline relies on a single pre-registered confirmation split.

VII. Conclusion

Weak persistent narrowband interference — the morphology that matters most for hydrogen-line spectrometry — is invisible to deep, classical, and industrial detector families alike, each for a different structural reason that injection maps make directly measurable. A label-free Fisher fusion whose members own the disjoint blind regions recovers it: with the official PatchCore stack as a calibrated fourth member, 0.997 AUROC on persistent narrowband, overall 0.914 [0.898, 0.929] on held-out images with CI separation from the strongest classical baseline, blind comb discovery and injection recovery on a real EUR 200 instrument, at $7.5\times$ the speed of the classical pipeline. The protocol — stratified injection maps, cluster-bootstrap CIs, selection/confirmation discipline — is instrument-agnostic and released for reuse.

Acknowledgment

The author thanks the RFI-NLN authors for the public LOFAR dataset and labelled masks.

Code and Data Availability

All experiments run on public data: the LOFAR and HERA benchmark (Zenodo record 8275061), the LAB survey cube, and the community corpora listed in the paper, each with their original licences. The detector suite (MRD, denoising AE, SumThreshold reference implementation, PatchCore/PaDiM arms, fusion, evaluation protocol), the reduced-scale runner for the original RFI-NLN code (with the four disclosed compatibility patches), the anomalib export harness, and every result file behind each table and figure (JSON with machine-readable provenance) are archived at `[repository DOI upon acceptance]`. The RSP1 raw IQ files are available from the author on reasonable request (instrument-owner data).

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